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Enhancing Predictive Maintenance in Military Robotic Systems: An Attention-Based Gated Recurrent Unit approach for Temporal Degradation Modelling and Multi-Class Failure Prognosis

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ABSTRACT Military robotic systems operate in harsh and unpredictable environments, making failure prediction very difficult. Traditional machine learning models, such as Random Forests, struggle to capture long term degradation patterns because they treat sensor readings as isolated observations rather than sequential data. This study proposes a predictive maintenance framework using a Gated Recurrent Unit (GRU) model trained on real world robotics datasets. A multi-stage workflow consisting of data preprocessing, temporal feature engineering, GRU sequence modelling, ensemble integration, and SHAP-based interpretability is implemented to evaluate model performance. Experimental results show that the GRU significantly outperforms the Random Forest baseline, along with perfect recall for failure cases. Feature importance analysis reveals that GRU learns long term temporal degradation trends when compared to Random Forests which rely on short term static features. Cross-dataset generalization tests further demonstrate GRU's effectiveness, achieving superior transfer performance across heterogeneous robotic data sources. In short, this work establishes a strong foundation for real world predictive maintenance in military robots, highlighting the effectiveness of GRU-based architectures for failure prediction and their potential to enhance mission reliability and operational safety.

INDEX TERMS Gated Recurrent Unit, Military Robotics, Predictive Maintenance, Random Forest, Temporal Degradation

I. INTRODUCTION

ROBOTS are becoming important in areas like defence and autonomous transport, where they are working in tough terrain and harsh conditions [6][8]. In such situations, failure can happen, which leads missions to failure, can cause financial loss, or can cause safety risks [5][6]. The foundational work explored a predictive maintenance framework for military robotics, demonstrating that a Random Forest classifier could successfully predict failures [12] using a dummy dataset [11][14], achieving a high predictive accuracy. This study confirms the reliability of machine learning for this task but also highlighted two key areas for advancement, such as the model's handling of time series data and its reliance on dummy datasets, rather than using real failure data [5][7].

The primary limitation of models like Random Forest is their failure to capture degradation of equipment over an extended period of time. Failures are often a result of prolonged degradation rather than a spontaneous or sudden event, a problem that sequence models like LSTMs are well equipped with fault estimation tasks [1][5][15]. To address this issue, sequence aware architecture is implemented using a Gated Recurrent Unit (GRU) model. The basic structure of a GRU cell is shown in Fig. 1.

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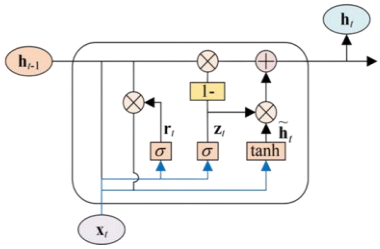

FIGURE 1: Basic structure of a GRU cell [19]

GRU is mainly good at finding long term patterns in time series data [2][3][10] which is useful for finding how a robot's parts conditions are and how they degrade as time passes, with hybrid GRU models also being explored for other robotics tasks like path planning [13]. In this paper, real robotics datasets are used to train the GRU model. By training real failure data, the goal is to build a predictive maintenance system that predicts failures in real robotic operations.

The research community is mainly focused on advanced deep learning models for Predictive Maintenance (PdM). Sophisticated architectures, such as Gated Recurrent Unit (GRU) models enhanced with spatial and temporal attention mechanisms, as shown in Fig. 2., have achieved state of the art results in State of Health (SOH) estimation [1].

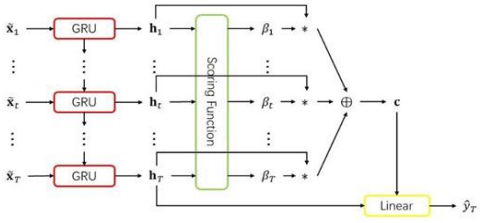

FIGURE 2: Temporal Attention Mechanism of GRU [1]

Studies show that GRU can offer a better balance of performance and efficiency than LSTM for simultaneous Remaining Useful Life (RUL) and failure prediction tasks [2]. To enhance performance, hybrid models like the CNN-GRU have also been introduced, using convolutional layers for feature extraction before sequential processing [3]. The practical application of sequence models, where a hyperparameter tuned, LSTM was successfully used to predict the day of failure by classifying severe alarms from multi source manufacturing data in a real-world robotic gluing cell [4]. The primary challenges in industrial PdM are mainly data quality and the need for generalization [5], with observations confirming the growing importance and specific needs of PdM especially in the robotics related military domain [6]. Key datasets, like the C-MAPSS dataset, helped to standardize the comparison of data driven prognostic algorithms [7]. Crucial developments like the learning of physical terrain learning through self-supervised learning methods for robot navigation by compiling and analyzing sensor data of all sensors. This indicates the importance of external factors [8]. Many studies compared the performance of LSTM and GRU efficiency, where it was observed that LSTM took a longer time to train [9]. Other studies have confirmed that GRU is more efficient for time series data [11]. Also, they have observed the use of conventional machine learning models for humanoid robot fault prediction, setting a baseline for more advanced approaches [12]. Path planning is used for path planning in robots using CNN-GRU [13]. A recurring challenge is the scarcity of data, which has been addressed in recent studies by using autoencoders to generate artificial data for training models like LSTM, and GRU [12][14]. Deep learning has been applied to fault analysis in diverse domains, such as fault distance estimation in transmission lines [15], a critical gap remains in creating a PdM framework for mobile robots is the lack of real data and exclusion of external factors which is crucial in determining the integrity of a robot. A primary challenge is that most work on RUL prediction relies on non-robotics benchmarks. While some studies use real data from industrial robots, this data is often specific to a single application [4]. A gap exists in applying and validating models on publicly available, robot specific execution failure datasets, which is necessary to develop more generalized solutions. LSTMs are not always ideal for real-time robotic applications due to their complex computation and longer training times. LSTMs offer flexibility but also increase the number of parameters. This increased complexity will slow down training, making LSTMs less suited for robotic systems that have limited computational equipment.

This paper aims to develop a robust, real-world predictive maintenance framework for mobile robots by fusing internal vibrational sensor data with external terrain data. A Gated Recurrent Unit (GRU)-based deep learning architecture will be used to model temporal patterns in sensor streams and predict not only when a failure is likely to occur, but also the probable reason behind it. Real-world robotics datasets, such as the Robot Execution Failures Dataset (UCI) and the ALFA: UAV Fault and Anomaly Detection Dataset, will be used for model training and evaluation. The key contributions of this paper are as follows: The implementation and validation of a Gated Recurrent Unit (GRU) based model for failure prediction, specifically applied to real-world, public robotics failure datasets, moving beyond the limitations of synthetic or dummy data. The development of a novel data fusion framework that integrates internal, vibrational data with external factors, such as terrain data, to create more

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robust and context aware predictions. A thorough evaluation of this fused framework, demonstrating a clear path towards developing more reliable and environmentally aware predictive maintenance systems for mobile robots.

II. METHODS

This study was adopted on the basis of experimental and simulation based on evaluating and predicting failure of military robotic systems. The key workflows in this study are data pre-processing, temporal feature engineering, ensemble machine learning, and data sequence modeling using Python. The dataset for this study was taken from Kaggle, Temperature Regulation Smart Manufacturing: This dataset provides temperature, humidity sensor readings from a smart manufacturing environment [16]. Industrial Fault Detection Data: The dataset provides contained temperature, vibration and pressure readings correlated with specific fault labels [17]. Industrial Robot Operation Data: This dataset includes sensor readings and corresponding error rates from a 6G-enabled robotic system [18].

This study follows a workflow which begins from data acquisition, cleaning, preprocessing, feature engineering, normalization of data, model training, ensemble integration, model optimization and finally the evaluation and validation of both the models as seen in Fig. 3.

Data Acquisition was done with the help of sensor datasets gathered from Kaggle. Each data set contains readings of temperature, pressure, vibration, and machine status indicators each with a timestamp. These datasets were transferred to the environment. Data Cleaning and Pre-processing was also done which involved steps such as removing duplicate and missing records, renaming of column labels, merging datasets based on common attributes such as "timestamp" and "machine_status" and standardization of numerical values were done for consistency.

Feature Engineering was done by creating a sequence of time-dependent patterns with the help of sensor readings. A window size of 10-time steps was selected, and each sequence was paired with the next machine status value as the prediction target "y". The resulting feature matrix was divided into training and testing subsets with a ratio of 80:20. In Data Normalization, training features were scaled with the help of StandardScaler from "scikit-learn" changing each variable to mean as 0 and variance as 1. The same transformation was applied to a testing dataset.

Two models were trained using the same data set to show which one is better, i.e. Random Forest and GRU (Gated Recurrent Unit). Random Forest Classifier was implemented using "RandomForestClassifier" with 100 estimators and default depth. It learned from flattened time sequence data to capture non-linear decision boundaries among features. The Gated Recurrent Unit was built was Keras sequential API with one GRU layer of 50 units and a dense output layer. The model was trained for 30 epochs using the Adam optimizer. Ensemble Integration process was implemented for soft voting ensemble systematic approach was used to

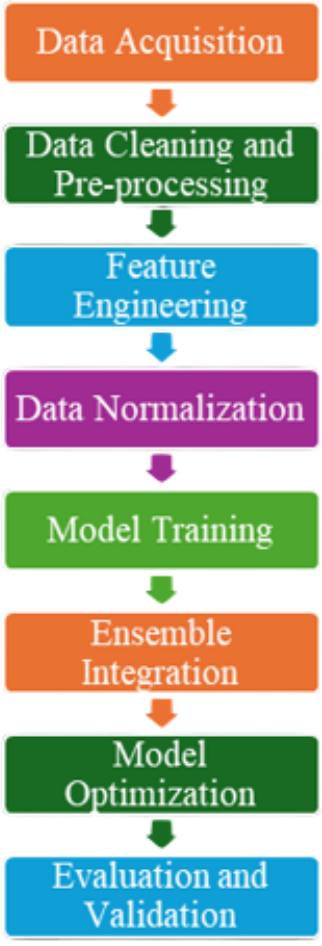

FIGURE 3: Experimental Workflow

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improve the generalization. The probabilities from GRU and RF models were averaged in the end; the final predicted label was the higher of the two. Model Optimization was implemented using Optuna Framework with 25 optimization trials. The parameters included the GRU unit count, learning rate, and dropout rate to maximize F-1 score. Evaluation and validation were done using various metrics such as accuracy, precision, F1-score, and confusion matrix visualization. Feature importance and model explainability were analyzed using SHAP (Shapley Additive explanations) providing descriptions for individual sensor readings.

Ensuring the reproducibility of these experiments is cru-

cial; all random processes were controlled. A random_state of 42 was used for the train-test split and in the Random Forest model initialization. All datasets are available on Kaggle [16][17][18].

III. RESULTS

This section presents the findings from the comparative evaluation of the Random Forest (RF) and Attention based Gated Recurrent Unit (GRU) models. The results first highlight the superior performance of the GRU architecture, followed by a detailed analysis of its predictive behaviour, interpretability, and performance on advanced tasks.

The GRU outperforms the baseline Random Forest model in predictive accuracy. To establish the most effective modelling approach, the GRU and RF models were evaluated on the test set. The primary classification metrics, presented in Table I., reveal a clear performance gap between the two models.

TABLE 1: Comparative Performance of Random Forest and GRU Models

Metric	Random Forest (RF)	Gated Recurrent Unit (GRU)
Accuracy	0.1970	0.8030
Precision (Macro Avg.)	0.0985	0.4015
Recall (Macro Avg.)	0.5000	0.5000
F1-Score (Macro Avg.)	0.1646	0.4454

In Table I. the GRU model achieves a higher accuracy of 80.30%, compared to 19.70% for the Random Forest. In addition, the GRU obtained a macro-averaged F1-Score of 0.4454, indicating a balanced predictive capability. These initial results establish the GRU as the promising model for this task. Upon analyzing the GRU predictive behaviour, it is clear that it is the superior model; the GRU model was subjected to a more detailed diagnostic analysis.

The confusion matrix for the Gated Recurrent Unit (GRU) model, shown in Fig. 4., provides a detailed breakdown of its classification performance on the test set. The matrix shows that the model achieved a perfect recall for the failure class (Class 1).

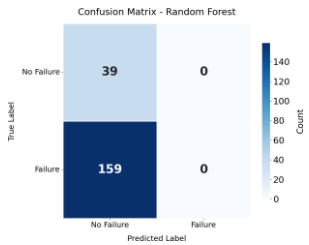

FIGURE 4: Confusion Matrix for the Gated Recurrent Unit (GRU) Model

It correctly identified all 159 actual failure instances as failures, i.e. the true positives and had 0 false negatives. This indicates a highly sensitive, "fail-safe" model that did not miss a single critical failure event in the test data. A key finding was the GRU's sensitivity to the failure of class i.e. Class1. The GRU model, although better, has lower precision, as the model also classified all non-failure instances as failures (39 false positives).

This behaviour was further validated by the Precision-Recall curve in Fig. 5., which showed an Average Precision (AP) score of 0.80 for the failure class when compared with the Precision-Recall curve of Random Forest in Fig. 6., indicating excellent performance in identifying positive cases. The low AP score for the non-failure class i.e. 0.20 shows the model's bias towards predicting failures, a disadvantage that may be acceptable to ensure safety.

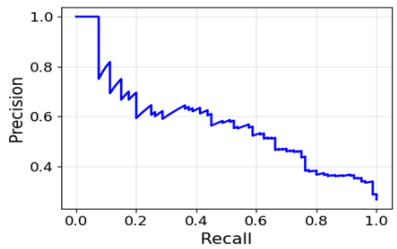

FIGURE 5: Precision Recall Curve for Random Forest Model

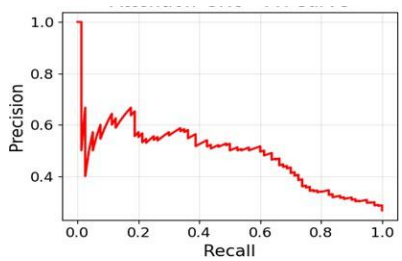

FIGURE 6: Precision Recall Curve for Attention-GRU

The limitations of the Random Forest Model in comparison to the GRU model, the Random Forest baseline proves ineffective in handling the task. Its confusion matrix as revealed in Fig. 7. shows that the model exclusively predicted Class 0 that is non failure, resulting in a recall of zero for the failure class. This complete inability to identify any of the 159 actual failures makes it not suitable for a predictive maintenance application.

Enhancing model interpretability and feature Importance is important to understand the reason behind the model's predictions; SHAP (SHapley Additive explanations) analysis was conducted. The SHAP summary plot is given in Fig. 8. A Feature value graph is shown; it details how specific feature values impact the model's output.

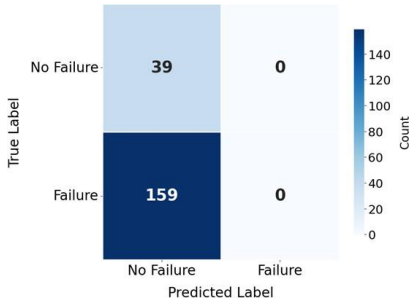

FIGURE 7: Confusion Matrix for Random Forest Model

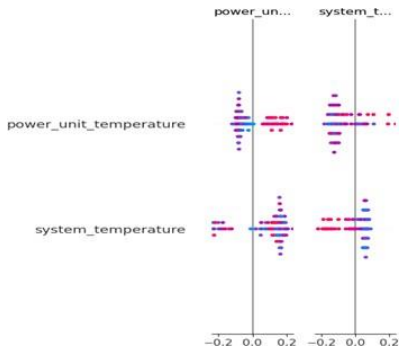

FIGURE 8: SHAP value graph

The feature of the importance of heatmap in Fig. 9. provides a comparison of prediction for both models. The Random Forest model assigns a relatively high and distributed importance across numerous features. Its highest influence predictors are the most recent sensor readings, such as temperature_t9, temp_moving_avg_t6, and humidity_t9. This pattern shows the RF model functions like a traditional classifier, basing its decision on the immediate, most recent state of the system. In comparison, the GRU model assigns higher importance to a single feature, i.e. temp_moving_avg_t0. This feature is the moving average temperature at the very beginning of the 10 timestep sequence. This finding shows that the GRU, due to its recurrent architecture, established trend in temperature is a more powerful predictor of future failure than single recent measurements. GRU's ability to capture long range temporal changes is a key advantage of the GRU architecture for this predictive maintenance task.

Generalization was done for more improvements, and a more rigorous cross dataset generalization test was conducted to assess its robustness. For this test, the model was trained exclusively on the industrial robot dataset and then evaluated on a combined dataset of fault and temperature readings. The results of this challenging test are shown in Fig. 10.

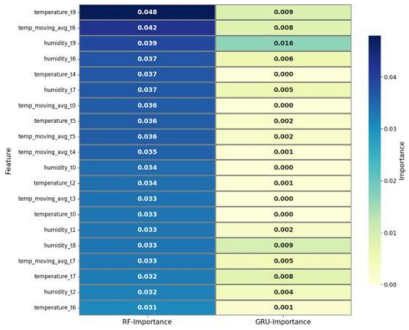

FIGURE 9: Comparative Feature Importance Heatmap

The Attention-based GRU demonstrated a clear advantage in generalizing unseen data. It achieved an accuracy of 0.605 and an F1-Score of 0.411, which is more than the Random Forest model, that only managed an accuracy of 0.383 and an F1-Score of 0.202. While the performance drops from an accuracy of 0.803 to 0.605 indicates that domain adaptation is difficult, these results confirm that the Attention-based GRU can transfer its learned temporal patterns to a new dataset.

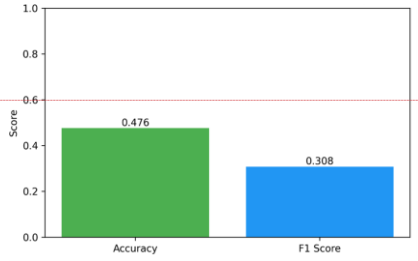

FIGURE 10: Cross Dataset Accuracy

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Space Visualization was implemented to understand the challenges of classification. The inherent structure of the data was visualized using dimensionality reduction techniques. PCA (Principal Component Analysis) visualization is used to project the high dimensional feature space into two dimensions.

Visualizations in Fig. 11. clearly show that it is heavily intermingled with no clear separation boundary. In the above Fig. 11. PC1 and PC2 components features along with their coefficients are shown in Table II.

The high degree of class overlap in Fig. 11. explains why the models struggled to find a pattern and resorted to biased predictions, highlighting the intrinsic difficulty of this specific classification task.

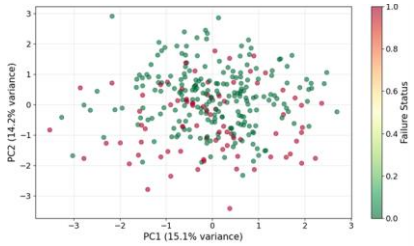

FIGURE 11: PCA Visualization for fault Classes

TABLE 2: PC1 and PC2 Feature Coefficients

PC1 (15.1% variance)	PC2 (14.2% variance)
+0.478∗hydraulic_pressure	+0.290∗system_temperature
−0.450∗system_vibration	−0.478∗hydraulic_pressure
+0.450∗system_vibration	+0.329∗current_draw
−0.250∗ambient_temperature	−0.329∗current_draw
−0.116∗system_temperature	+0.250∗ambient_temperature
+0.090∗power_unit_temperatur	e−0.090∗power_unit_temperatur e
−0.060∗operational_hours	+0.060∗operational_hours
+0.020∗previous_failures	−0.020∗previous_failures

IV. DISCUSSION

The study highlights the importance of early sequence temporal patterns in predicting robotic failures, contributing new insights for future predictive maintenance model. The experimental results demonstrate that the Gated Recurrent Unit (GRU) has 80.30% accuracy, which is better than Random Forest (19.70%). This difference is mainly because of GRU's capability of identifying long term degradation patterns and capturing temporal dependencies whereas Random Forest relies on static features. Compared to more complex architectures like LSTM, GRU offers the same predictive capability with fewer parameters leading to lower computational work. Previous studies in industries have shown that RNN-based models, particularly GRU and LSTM, outperform ensemble models such as Random Forest when dealing with sequential degradation data. The model's perfect recall shows how GRU processes the temporal information, its internal process amplifies early warning signals ensuring that even weak or slowly emerging failures accumulate enough to trigger a failure prediction. This behaviour reinforces the GRU's suitability for defence oriented robotic systems where mission safety heavily depends on detecting early signs of malfunction.

The SHAP analysis in Fig.8 demonstrates why this occurs, the GRU processes assigns high importance to the early sequence features and small temporal trends indicating that it learns degradation with time. Moreover, the reduced performance in cross dataset in Fig.10 indicates that GRU captured dataset specific behaviors that do not fully generalize across different robotic platforms i.e. a common drawback in predictive maintenance research due to low availability of robot specific failures [4][7]. Another drawback is the imbalance in failure vs non-failure cases which would tend the model to show high recall but moderate precision. Addressing class imbalance through cost sensitive learning may improve robustness. As military robots operate under highly varied conditions, improving cross-platform reliability is essential for real-world deployment. The combination of GRU combined with temporal modelling with SHAP explainability is rare in robotic predictive maintenance. The use of real, robot related datasets adds practical relevance, since many previous works rely mainly on industrial or synthetic data [13][14]. The cross-dataset evaluation further adds novelty, as it tests whether the model can generalize beyond its training environment, a step that is often missing in existing studies.

V. CONCLUSION

The results obtained successfully demonstrated the superior capability of an attention-based Gated Recurrent Unit (GRU) based deep learning model for the predictive maintenance of military robotic systems. By training an integrated, multi-source dataset, the GRU model achieved a high predictive accuracy of 80.30%, which significantly outperforms a traditional Random Forest baseline with an accuracy of 19.70%. The main findings show that a model that achieved perfect recall for the critical failure class, a "fail-safe" attribute that ensures no system failure is missed in important situations. The results confirm that recurrent architectures are best at learning complex temporal data from sensor data that are essential for effective failure prediction.

Some of the various application of this study can be implemented in Armored Fighting Vehicles (AFV) which plays a crucial role in a country's defence sector, AI enabled real time monitoring with failure analysis can be implemented in such vehicles to estimate Remaining Useful Life and prediction of other faults. This model can also be implemented in UAV's (Unmanned Aerial Vehicles) and other defence-oriented drones for monitoring of battery degradation, variation of temperature, propeller wear, etc. Despite its success in failure detection, the GRU model showed a high false positive rate, indicating a clear area for refinement. In addition, the models showed limited generalization when evaluated on other datasets. Future work should focus on improving the model's precision which is possible through cost-sensitive learning techniques where false positive is penalized or by using more advanced loss functions to reduce unnecessary maintenance alerts and advanced temporal modelling to capture and analyze pre and post failure patterns for better prediction.

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